

Milestone 3: Documentation

Maintenance, Cost Management & Analytics

Objective

The objective of Milestone 3 is to extend the laboratory resource management platform with maintenance scheduling, work order management, calibration tracking, certification renewal reminders, cost tracking, inter-institution billing, analytics dashboards, utilization and cost reports, and notification and alert mechanisms. The milestone focuses on improving equipment lifecycle management, financial visibility, operational monitoring, and timely communication within the laboratory environment.

Scope

- Maintenance scheduling and work order management
- Equipment calibration tracking
- Certification expiry and renewal reminders
- Equipment and operational cost tracking
- Inter-institution billing workflows
- Analytics dashboards for researchers, lab managers, and administrators
- Utilization effectiveness and cost analysis reports
- Notification and alert systems

Functional Requirements

1. Maintenance Scheduling & Work Order Management: Schedule preventive and corrective maintenance activities, maintain work orders, record maintenance status and support the workflow from maintenance request through completion.
2. Calibration Tracking & Certification Renewal: Maintain calibration information for equipment, record calibration dates and due dates, track certification validity, and provide reminders when calibration or certification renewal is approaching.
3. Cost Tracking & Inter-Institution Billing: Record relevant equipment and maintenance costs, support cost visibility for resource usage, and provide workflows for billing when laboratory resources are used across institutions.

4. Analytics Dashboards: Provide role-oriented analytics views for researchers, lab managers, and administrators. Dashboards should present meaningful utilization, maintenance, cost, and resource information to support operational decisions.

5. Utilization Effectiveness & Cost Analysis Reports: Generate reports that combine resource utilization and cost information. The reports should help users evaluate equipment effectiveness, identify usage patterns, understand associated costs, and support resource planning.

6. Notification & Alert System: Generate timely notifications and alerts for events such as upcoming maintenance, calibration due dates, certification expiry, billing events, work-order updates, and other important platform activities.

Frontend Modules

- Maintenance Scheduling Page
- Work Order Management Page
- Calibration Tracking Page
- Certification Renewal / Expiry Page
- Cost Management Page
- Inter-Institution Billing Page
- Analytics Dashboard
- Utilization & Cost Reports Page
- Notifications and Alerts Page

Backend Modules

- Maintenance Scheduling Service
- Work Order Service
- Calibration Tracking Service
- Certification Management Service
- Cost Tracking Service
- Inter-Institution Billing Service

- Analytics Service
- Reporting Service
- Notification and Alert Service

Core Workflow

The milestone connects operational and analytical workflows. Equipment can be scheduled for maintenance and represented through work orders. Calibration and certification dates can be tracked independently so that upcoming deadlines can trigger reminders. Cost data can be associated with equipment usage and maintenance activities, while inter-institution usage can be processed through billing workflows. The resulting operational data can then be presented through dashboards and reports, with notifications keeping relevant users informed about important events.

Database Tables / Data Areas

- Equipment — equipment identity and lifecycle information
- Maintenance Records — maintenance activity and history
- Work Orders — maintenance requests, assignments, status, and completion information
- Calibration Records — calibration dates, due dates, and related records
- Certification Records — certification details and expiry information
- Cost Records — equipment, maintenance, and usage-related costs
- Billing Records — inter-institution billing information
- Utilization Records — equipment usage information used for effectiveness analysis
- Notifications — reminders, alerts, and system notifications

Technologies

- Frontend: React.js, React Router, Axios, Tailwind CSS, Vite
- Backend: Java, Spring Boot, Spring Security, REST APIs, Hibernate/JPA, Maven
- Database: PostgreSQL

- Development & Testing Tools: GitHub, Visual Studio Code, IntelliJ IDEA, Postman, pgAdmin 4
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Expected Deliverables

Maintenance scheduling and work order management workflows; calibration tracking and certification renewal reminders; cost tracking and inter-institution billing workflows; analytics dashboards for researchers, lab managers, and administrators; utilization effectiveness and cost analysis reports; and notification and alert systems.

Learning Outcomes

Develop an understanding of maintenance and calibration management workflows, equipment lifecycle management, cost tracking and billing processes, analytics dashboard design, utilization and cost analysis, report generation, and notification-driven resource management. The milestone also supports integration of operational data with analytical and decision-support features.

Conclusion

Milestone 3 extends the laboratory resource management platform beyond booking and utilization by introducing maintenance, calibration, certification, cost, billing, analytics, reporting, and notification capabilities. These features provide a more complete equipment lifecycle and resource management ecosystem while helping researchers, lab managers, and administrators make informed decisions about equipment availability, maintenance, utilization, and cost.

